

Forecasting Consumer Sentiment Index

Using Google Trends and Macroeconomic Data

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1 Introduction

The University of Michigan Consumer Sentiment Index (CSI) is a survey that measures how consumers feel about their future financial situation. CSI is a popular economic indicator that serves as a proxy estimate of future economic growth. Implicitly, the more confident that consumers are in the economy, their job, and their income, the more likely they are to spend money in the economy. **Our project intends predict the United States Consumer Sentiment Index (CSI) using both Google search trends and economic indicators, with the intent of understanding underlying factors of a consumer's confidence in the economy and to give highly precise CSI estimates.** The Consumer Sentiment Index is a measurement of human behavior and how consumers feel about the economy, not necessarily how the current economic condition is from an objective perspective. The metric of Consumer Sentiment is highly relevant to human behavior as a reaction to economic and financial conditions. Hence, we believe that with the right selection of keywords in Google search trends and macroeconomic indicators, we can leverage time series regression models such as Autoregressive Integrated Moving Average (ARIMA), Vector Auto Regression (VAR), and Mixed Data Sampling (MIDAS) models to extract the underlying sentiment of the average consumer on their current and future economic position. While others (Choi H. (n.d.)) have worked in similar designs before, this will be the first model to use both Google Trends and Economic Indicators in the United States with contemporary keywords selection. We developed a highly accurate model

using VAR and 6 predictors with an R^2 of 0.972 and RSE of 2.08 for 2020 to 2024 testing. This paper will first provide an overview for other work, then dive into how we extract the data and developed the model. Then we will discuss their performance and some potential ethical concerns.

2 Related Work

In exploring methodology to ultimately predict CSI, our research has been informed by a selection of literature, closely related to our approach. Three research stood out in providing guidance for our model, those being “Using Machine Learning Methods to Predict Consumer Confidence”, “Consumer Sentiment with Google Trends”, and “Predicting the Present with Google Trends”.

In “Using Machine Learning Methods to Predict Consumer Confidence”, Han, Li, and Li (2023) present an analysis of the Consumer Confidence Index (CCI) in China, utilizing a variety of machine learning and deep learning techniques, including several neural network architectures and Random Forest, with data sourced from search engines. This study proved valuable in offering a diverse set of computational methods and in suggesting important search term categories that might correlate with consumer sentiment metrics. While CCI and CSI differ—CCI being more focused on economic confidence and CSI on purchasing attitudes towards household goods—the overlap between them has informed our selection of relevant search terms for our predictive model.

“Consumer Sentiment with Google Trends” was a Google study that constructed a CSI prediction model utilizing Google Trends data. In said piece, [Choi H. \(n.d.\)](#) split Google Search Trends data into categories corresponding to the five underlying questions of the CSI. By identifying search queries with strong correlations to the time series data of each question, they were able to regress these trends to forecast the CSI, potentially employing autoregressive models. This approach parallels our utilization of Google Trends for CSI prediction. However, our model is distinct in its use of varying regression techniques and the integration of macroeconomic data to enrich the analysis.

Lastly, we examined a paper “Predicting the Present with Google Trends”, that leveraged Google Trends data for real-time (“nowcasting”) predictions of various economic indicators. In this work, [Choi and Varian](#) demonstrated the potential for timely estimations of economic conditions such as unemployment rates, tourism statistics, and general consumer confidence by carefully curating categories provided by Google Trends. While there are similarities to our approach in terms of employing time series regression and the predictive use of Google Trends, our methodology diverges by incorporating macroeconomic variables and favoring Vector Autoregression, MIDAS regression over simpler autoregressive models to enhance the accuracy and robustness of our predictions.

These studies reviewed have provided foundational insights that guide our approach to predicting CSI; they underscore the significance of selecting pertinent search terms, the value of incorporating real-time search data, and the need for sophisticated modeling techniques to account for the multifaceted nature of economic indicators. Our model aims to utilize these learnings with our own innovations to produce a reliable predictive tool for CSI.

3 Data

Our data come from three different sources: Google Trends, Federal Reserve Economic Data (FRED), and Yahoo Finance. They were collected in the first week of April 2024 from 2004 to 2024. A total of 60 variables were collected, and 3 were used in the final Vector Auto Regression model. Table 1 provides an overview of all variables tested. We selected these variables based on the

questions of the CSI survey and prior research works. We also selected these variables based on the understanding of the domains.

Sources	Variables Tested
Google Trends	“inflation”, “unemployment”, “interest rate”, “rental housing”, “job opening”, “economy”, “tourism”, “layoff”, “employment”, “debt management”, “rent”, “wages”, “real estate”, “used car”, “gas price”, “recession”, “housing market”, “tax cut”, “federal reserve”
FRED Monthly	Total Healthcare Construction Spending, One-Year Real Interest Rate, Two Year Expected Inflation Rate, Inventory to Sales Ratio, Total Utility Capacity Index, AP Grade A Egg Price, 12-Month Moving Average of Unweighted Median Hourly Wage Growth: Age: 16-24 Years, Inflation Expectation, 30-Day AA Financial Commercial Paper Interest Rate, 60-Day AA Financial Commercial Paper Interest Rate, 90-Day AA Financial Commercial Paper Interest Rate, 30-Day AA Nonfinancial Commercial Paper Interest Rate, 60-Day AA Nonfinancial Commercial Paper Interest Rate, AD&Co US Mortgage High Yield Index, Credit-and-Option-Adjusted Spread: Mid-Tier, AD&Co US Mortgage High Yield Index, Credit-and-Option-Adjusted Spread: Tier 0 (1,2,3), 12-Month Moving Average of Unweighted Median Hourly Wage Growth: Wage Distribution: 1st to 25th (25th-50th, 50th-75th, 75th-100th) Wage Percentile, Total Vehicle Sales, 3-Month Median Salary, AD&Co US Mortgage High Yield Index: Tier 0 (1)
FRED Weekly	Federal Funds Rate, US Regular Conventional Gas Price, US Regular Reformulated Gas Price, Insured Unemployment Rate, Chicago Fed National Financial Conditions Leverage Subindex, St. Louis Fed Financial Stress Index
Yahoo Finance	S&P 500, CBOE VIX

Table 1: Overview of All Tested Variables

*parentheses contain extra variables

We utilized different techniques to extract information from various sources. Google Trends limits the download of keywords to 5 for each request and performs a normalization between them to make it more comparable for use. However, this process can make the data less useful for machine learning algorithms. A denormalization process was performed using the R package *gtrendsR* with anchor terms “white,”

“blue,” “coca cola,” “seaweed,” and “perspicacious,” which remain relatively constant on Google Trends over time. These variables provide a reference to denormalize data and make them more comparable and approximate the absolute interests.

For Yahoo Finance data, we calculated the absolute difference between each trading period, defined as a week or month. We used the simple calculation of close - opening to gather information on the fluctuation of the market. We did not perform additional manipulation on FRED data.

4 Methods

We employed various techniques to select the best variables for our model. Two statistical tests were performed, in addition to a visual analysis.

4.1 Visual Analysis

We performed a visual analysis for variable itself by plotting CSI with each variable of interest to qualitatively understand if a potential variable would be useful for making predictions. Some variables were removed from this test due to confounding factors from other events. For example, the outbreak of the COVID-19 influences the “employment” keyword and unemployment rate. “gas price” was also influenced by the outbreak of the Ukraine war. Using the domain knowledge combining the relevant news we removed the candidates that had significant confounding influences.

The ideal candidate will show an either negative or positive correlation with the CSI variable. As demonstrated by Figure 1. To quantify this impact, we performed the cross-correlation test.

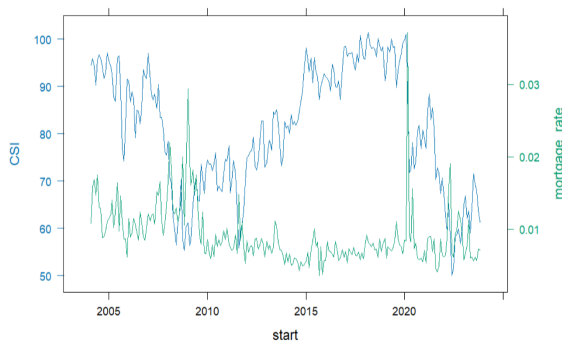


Figure 1: Visual Analysis of CSI and Mortgage Rate

4.2 Cross-Correlation Test

We used the sample cross-correlation function (CCF) to identify correlating lags of our candidate variables to CSI. We set the maximum lag to 30 to be as exhaustive as possible. We operated this after understanding the stationarities of the candidates, i.e., the mean, variance, and autocorrelation structure do not change over time. The CCF function will output an array of values that are between -1.0 and 1.0, where an ideal candidate will have a value closer to the boundary and has a negative association when closer to lag 0. This is an indication of a lagging variable having the potential of predicting a future variable, which is consistent with what can be done in real life (using past data to predict the future data) Figure 2 shows the “mortgage-rate” from Google Trends that has a potential of predicting the CSI and serves as a reference of what an ideal candidate looks like.

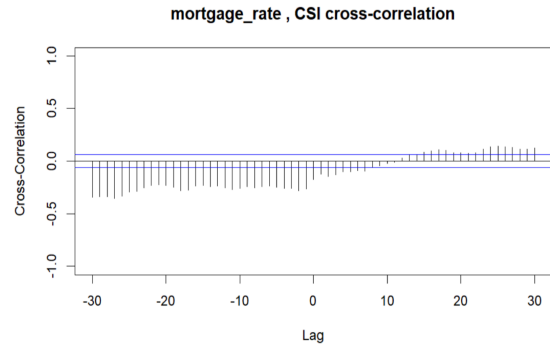


Figure 2: CCF Result of Mortgage Rate and CSI

4.3 Granger’s Causality Test

Granger’s Causality is a test to determine if a candidate has the potential of being used to forecast a time series. This statistical test uses the prior values of the candidate time series to test the causal effects on the future CSI values. We used lags = 1, 2, 3 for the purpose of testing lags and obtained the F-statistics by comparing these two time series.

For null hypothesis:

$$y_t = a_0 + a_1 y_{t-1} + a_2 y_{t-2} + \dots + a_m y_{t-m} + \varepsilon_t \quad (1)$$

For alternative hypothesis:

$$y_t = a_0 + a_1 y_{t-1} + a_2 y_{t-2} + \dots + a_m y_{t-m} + b_p x_{t-p} + \dots + b_q x_{t-q} + \varepsilon_t \quad (2)$$

The alternative hypothesis includes augmented variables from the candidate series, where p stands for the nearest lag and q stands for the furthest lag.

We set the p-value threshold to 0.05, and Table 2 lists the variables we found to be statistically significant.

Variable	Mean p-value
Mortgage Rate (Google Trend)	0.002
Bankruptcy (Google Trend)	0.019
Employment (Google Trend)	0.023
Used Car (Google Trend)	0.026
Rental Housing (Google Trend)	0.029
Total Vehicle Sales (FRED)	0.003

Table 2: Granger’s Causality Test

5 Algorithms

We constructed 3 different models for our data: an ARIMA baseline model, a VAR model for monthly data, and a MIDAS model for weekly data. We will present the methodology as well as the results below. We plan to use R^2 and RSE to evaluate our model. All models are trained follow strict test/train split temporal separation to prevent future data leaking to the model. Training are performed using data from 2004-2020 and all models are evaluated from 2020 to 2024.

5.1 ARIMA Model

The first model that we used as our baseline model for univariate time series data is the Autoregressive Integrated Moving Average (ARIMA). ARIMA is particularly effective in capturing the patterns and inherent structures in historical data to predict future values. The ARIMA model is characterized by three parameters: (p, d, q) , where p is the number of autoregressive terms, d is the number of differences needed to make the series stationary, and q is the number of lagged forecast errors in the prediction equation.

For the analysis of CSI, we selected an ARIMA (2,0,2) based on maximizing Akaike’s Information Criterion (AIC), while also keeping the model relatively small to ensure numerical stability when estimating its parameters.

The ARIMA (2,0,2) model was selected as the baseline for our study for several reasons. First, as a relatively simple model, it facilitates

understanding and interpretation of our parameters, which is crucial for an initial exploratory analysis. Secondly, starting with a baseline ARIMA model allows for the establishment of a benchmark against which the performance of more complex models can be measured.

In evaluating our model, we employed a 1-step ahead forecasting system, where after each forecast is made, the model parameters are refit with all the previous data. This method of forecasting best simulates the monthly forecasting that would be made if this model were ultimately implemented for predictions. Figure 3 shows a plot of the forecasted data, along with 95% confidence intervals, as well as the ground truth data. The model achieved Residual Standard Error of 5.31 and an R^2 value of 0.810.

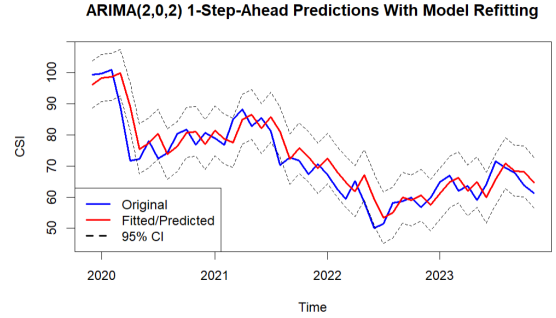


Figure 3: Forecast of ARIMA (2,0,2) with original data and 95% CI

5.2 Vector Autoregression (VAR) Model

We then moved on to Vector Autoregression (VAR) to capture the dependencies between CSI and our other predictive data. Vector Autoregression (VAR) is a multivariate model that is used to capture the linear interdependencies between multiple time series. The VAR model is a generalization of the univariate Autoregression

model that allows for more than one evolving variable.

Our VAR model is structured as follows, with regression results shown in Figure 4:

$$\begin{aligned} CSI = & CSI.l1 + employment.l1 \\ & + mortgageRate.l1 \\ & + totalVehicleSales.l1 \\ & + const + trend \end{aligned}$$

where *l1* indicates a one-month lag

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Estimation results for equation CSI:
=====
CSI = CSI.l1 + employment.l1 + mortgage_rate.l1 + totalVehicleSales.l1 + const + trend

              Estimate Std. Error t value Pr(>|t|)
CSI.l1         0.84112    0.04437  18.959  <2e-16 ***
employment.l1  -21.28986   20.48761  -1.039   0.3001
mortgage_rate.l1 -167.70215 103.55956  -1.619   0.1071
totalVehicleSales.l1 0.43823  0.22641   1.936   0.0545 .
const          12.32077    5.25187   2.346   0.0200 *
trend          -0.01353    0.01640  -0.825   0.4106
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.871 on 183 degrees of freedom
Multiple R-Squared: 0.9002, Adjusted R-squared: 0.8975
F-statistic: 330.3 on 5 and 183 DF, p-value: < 2.2e-16
```

Figure 4: Summary of VAR

Unsurprisingly, the lag of CSI had the lowest p-value of all our variables at $2e^{-16}$. The Google search trend of the term “mortgage rate” and data on US total vehicle sales were also highly predictive of CSI. We didn’t find that any of our variables had significance beyond 1 lag.

The model was able to forecast our data very effectively, using the same forecasting methodology as the ARIMA model. The model was able to achieve a Residual Standard Error of 2.08 and an R^2 of 0.972. Both metrics are vast improvements upon our baseline ARIMA model. Figure 5 displays the forecasting results.

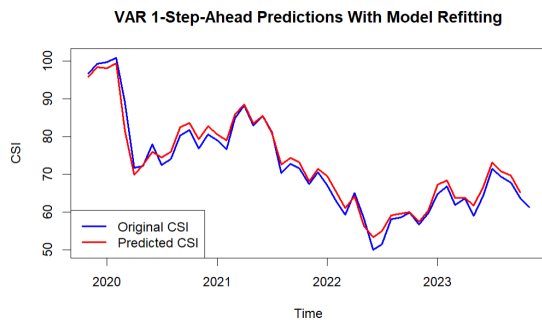


Figure 5: Forecast of VAR With Original Data

5.3 Mixed Data Sampling (MIDAS) Model

Although the VAR model was able to explain the data very well, we hoped to forecast weekly CSI, rather than monthly, to give a timelier forecast of this important metric. The Mixed Data Sampling (MIDAS) model is an econometric approach designed to handle datasets where observations

occur at different frequencies. This type of modeling is particularly useful in our case where we had weekly Google search trend and macroeconomic data as our predictive variables but monthly target variables in CSI. MIDAS allows high-frequency data to inform predictions of low-frequency outcomes without the need to aggregate the high-frequency data into lower frequencies, thus preserving much of the information in the data.

Below is the structure of our model:

$$\begin{aligned} CSI = & intercept + CSI.l4 + employment.l1 \\ & + employment.l2 \\ & + rentalHousing.l1 \\ & + rentalHousing.l2 \\ & + mortgageRate.l1 \\ & + mortgageRate.l2 \\ & + totalVehicleSales.l1 \end{aligned}$$

Unfortunately, the MIDAS model performed much worse than the VAR model and only slightly better than the baseline model, with a Residual Standard Error of 4.93 and an R^2 of 0.836. Figure 6 shows the forecast results on the test data, along with the ground truth data, forecasted with the same methodology as the previous two models.

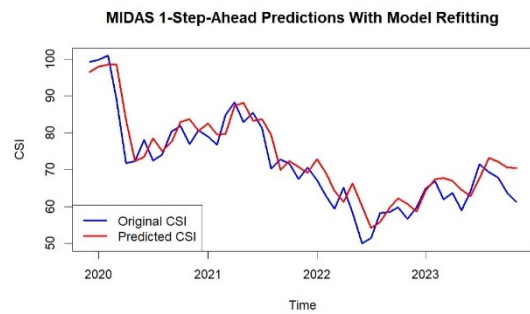


Figure 6: Forecast of MIDAS With Original Data

6 Results and Discussion

Our first key takeaway from our modeling process was the effectiveness and simplicity of the VAR model in forecasting monthly Consumer Sentiment Index. Using only 6 parameters, we were able to achieve an R^2 of 0.972 and an RSE of 2.08. A key reason for that is the feature selection of our variables that can effectively capture the changes in CSI. Additionally, VAR is very capable at capturing variable-temporal relationships and maximizing parameter likelihood.

Our second key takeaway from our modeling process was how poorly MIDAS performed relative to our simpler baseline ARIMA model and the better VAR model, with R^2 and RSE only slightly better than the ARIMA model. We have two hypotheses as to why this was the case. The first is that the parameter estimation methods for VAR may be more effective at maximizing the likelihood than that of MIDAS. VAR commonly relies on Ordinary Least Squares (OLS), whereas MIDAS often relies on Non-linear Least Squares or gradient based optimizations. Due to this, the parameter estimation methods for these respective models may lead to VAR providing more effectively maximized parameters than MIDAS. The second hypothesis for why MIDAS wasn't as successful is that there is less information contained in weekly data, especially weekly data that is only 1-2 weeks before the target date, than in monthly data from the previous month. This might lead to more information being captured more succinctly in the VAR modeling process than in MIDAS.

Model	ARIMA	VAR	MIDAS
RSE	5.31	2.08	4.39
R^2	0.81	0.973	0.836
Predictors	4	6	9
p-value	<2.2e-16	<2.2e-16	<2.2e-16

Table 3: Model Evaluation Results

7 Ethical Implications

7.1 Sampling Bias

Given that Google is used by most of the population, and our word selection seems to encompass the general US population, it would appear likely that our system will perform well in a heterogeneous population. However, the people who don't have internet access (approximately 7% of the adult population; via Pew Research), whether that be because they are from an older generation, have poor financial situations, or they live in rural areas, might be under-represented in our model. If that is the case, this could lead to bias in our model. We will further explore whether the sample of Google users is representative of the US population.

Furthermore, while Google commands a substantial 87.48% share of the search engine market in the U.S. (gs.statcounter.com), it is important to recognize that a significant number of users still opt for alternative search engines like Bing. The model may not fully capture the behaviors of those who prefer other search engines, which introduces a potential lack of representation.

7.2 Caveat Emptor

The consequences of inaccurate predictions in a CSI prediction model can be significant. If the system makes mistakes, individuals may form incorrect perceptions about the economy, which could influence major financial decisions, investment choices, and overall consumer behavior. Our model only uses previous CSI Values, Total US Car Sales, and the Google Search Trends of "mortgage rate" and "employment" so it isn't all encompassing of the economy and price of household items, so we strongly advise against using our model to make any major financial decisions, as its purpose is to be predictive and provide insight into what factors influence consumer sentiment.

8 Conclusion

Using Google Trends and macroeconomic data, we were able to effectively model future Consumer Sentiment Index values very effectively, capturing the underlying factors that motivate change in consumer sentiment on a national level. Without a doubt, our work was a success in its ability to forecast future values of CSI with exceedingly high accuracy. Vector Autoregression proved to be an effective statistical method to model the temporal structure of CSI. It was highly revealing that employment, mortgage rate, and total vehicle sales were highly predictive of CSI.

Additional work in the future can be focused on two areas. The first being additional feature selection to incorporate other variables (whether it be Google Trends, macroeconomic data, or other) to predict CSI that would add additional insight into the factors that affect CSI and that would further increase the model's predictive ability. The second area of future focus would be working with MIDAS and other mixed-frequency modeling methods that could forecast CSI on a weekly basis. It would be incredibly useful to a variety of bankers, politicians, manufacturers, and other

working positions to have forecasts of CSI at a higher frequency so they can make real-time decisions in their line of work with up-to-date forecasts.

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